

Chemo-mechanical failure of solid composite cathodes accelerated by high-strain anodes in all-solid-state batteries

Junhee Kang^a, Hong Rim Shin^b, Jonghyeok Yun^a, Siwon Kim^a, Beomsu Kim^a, Kyeongsu Lee^c, Youngjun Lim^c, Jong-Won Lee^{a,*}

^a Division of Materials Science and Engineering, Hanyang University, 222 Wangsimni-ro, Seongdong-gu, Seoul 04763, Republic of Korea

^b Department of Energy Science and Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), 333 Techno Jungang-daero, Hyeonpung-eup, Dalseong-gun, Daegu 42988, Republic of Korea

^c Next Generation Development Team, Samsung SDI Co., LTD., 130 Samsung-ro, Yeongtong-gu. Suwon-si, Gyeonggi-do 433-803, Republic of Korea

ARTICLE INFO

Keywords:

All-solid-state battery
Oxide cathode
Anode volume change
Interfacial heterogeneity
Chemo-mechanical degradation

ABSTRACT

Large volume changes in Li-based anodes during repeated charge-discharge cycling, which can exert additional mechanical stresses on cell components, remain a significant bottleneck for realizing all-solid-state batteries (ASSBs). While a few studies have reported the mechanical deformation of solid electrolyte layers induced by the volume changes in anodes, the possible degradation of composite cathodes has been largely overlooked. Herein, we present a comparative experimental-simulation study of sulfide-based ASSBs assembled with high-strain (Li-In) and zero-strain ($\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO)) anodes to understand the impact of anode volume changes on the chemo-mechanical degradation of composite cathodes. The Li-In cell suffers from severe capacity loss after ~ 120 cycles, whereas the LTO cell shows a capacity retention as high as 76 % over 200 cycles. In-depth chemical and microstructural analyses, coupled with impedance decoupling and mechanical simulations, reveal that the combination of the cathode volume changes and the high-strain Li-In anode perturbs the structural integrity of the $\text{LiNi}_{0.88}\text{Co}_{0.09}\text{Al}_{0.03}\text{O}_2$ (NCA) composite cathode and facilitates “dynamic” contacts among the cathode constituents upon repeated cycling. This leads to enhanced parasitic interfacial reactions, as evidenced by the increased amount of resistive phases in the cathode. The resulting chemically/electrochemically heterogeneous interfaces between the NCA and $\text{Li}_6\text{PS}_5\text{Cl}$ lead to accelerated cracking of the NCA aggregates in the presence of anode-induced stresses. This study highlights the accelerated degradation of composite cathodes driven by high-strain anodes and provides insights into the design of ASSBs with long cycle lifetimes.

1. Introduction

Lithium-ion batteries (LIBs) play a major role in energy storage for a wide range of portable electronic devices and electric vehicles [1–3]. However, the state-of-the-art LIBs are reaching their upper limit of energy density [4–6], which is not high enough to meet the ever-growing demand for large-scale, high-energy-density batteries. More importantly, safety issues arising from the flammable nature of liquid electrolytes remain a critical problem that must be addressed for large-scale batteries [7–9]. In this regard, all-solid-state batteries (ASSBs) with non-flammable solid electrolytes (SEs) have emerged as promising alternatives to LIBs, which are based on liquid electrolytes [10–14]. ASSBs not only reduce the risk of battery fires but also enable the utilization of high-capacity anodes (e.g., Li metal and alloys) [15–17] and bipolar

designs [18,19], which can boost both gravimetric and volumetric energy densities. Among various SE candidates for ASSBs, sulfide SEs (e.g., $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl), Li_3PS_4 , $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$) have been extensively studied, owing to their high ionic conductivities ($1\text{--}25\text{ mS cm}^{-1}$) at room temperature [20–22].

Highly dense and robust SE layers are expected to effectively suppress the formation and propagation of Li dendrites, thus enabling the utilization of various high-capacity anodes such as those based on Li metal and its alloys or Li-free Ag-C composites [23–25]. Recently, Lee et al. reported an “anodeless” (Li-free) ASSB using a carbon layer embedded with Ag nanoparticles, in which a Li alloy was formed between the carbon layer and current collector during charging [26]. However, both Li-based and Li-free anodes undergo large volume changes of 50–320 % during repeated Li plating and stripping,

* Corresponding author.

E-mail address: jongwonlee@hanyang.ac.kr (J.-W. Lee).

<https://doi.org/10.1016/j.ensm.2023.103049>

Received 3 August 2023; Received in revised form 1 November 2023; Accepted 4 November 2023

Available online 5 November 2023

2405-8297/© 2023 Elsevier B.V. All rights reserved.

depending on the material and composition [27,28]. High volume changes in anodes deteriorate their structural integrity, resulting in low Coulombic efficiency and poor cycling stability.

In addition to inducing morphological and mechanical deformation of anodes, massive volume changes exert additional stress on the cell components and their interfaces [29–32]. In conventional LIBs, liquid electrolytes can accommodate the breathing of electrodes, effectively convert mechanical strain into hydrostatic pressure, and ensure uniform distribution throughout the cell [33,34]. However, in the case of ASSBs under external pressure, any mechanical stress/strain evolution in the anode can trigger detrimental microstructural changes within the cells. For example, the compressive strain generated by anode volume changes can initiate plastic deformation of the SE layer [35,36]. Budiman et al. demonstrated that anode materials exhibiting a high expansion rate during battery charging (Li plating) induce 47 % higher maximum von Mises stress in the sulfide SE layer compared to low-strain anodes, eventually generating cracks in the SE layer [37]. Similar phenomena have been observed in Si-based anodes during charging [38]. Given the integrated structure of an ASSB cell, anode volume changes and the resultant mechanical instability can have a negative influence on the composite cathodes about which little is known.

In this study, we conducted an in-depth investigation of the degradation behavior of a polycrystalline $\text{LiNi}_{0.88}\text{Co}_{0.09}\text{Al}_{0.03}\text{O}_2$ (NCA) composite cathode in ASSBs accelerated by high-strain anode materials. A Li-In alloy was employed as a high-strain anode instead of Li metal owing to its electrochemical reversibility and stability with LPSCI [17,29], thereby allowing us to focus on anode volume changes. For comparison, we also utilized a $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) anode, which exhibits zero strain upon Li intercalation-deintercalation. The LTO cell demonstrated stable capacity retention over 200 cycles, whereas the Li-In cell exhibited rapid capacity decay after ~120 cycles, with significant signs of parasitic reactions and NCA particle cracking. The influence of anode volume changes on cathode degradation was studied in depth using quantitative impedance decoupling combined with post-mortem chemical and microstructural analyses. Furthermore, 2D electrochemical-mechanical simulations were performed to investigate the influence of the anode-induced stress on the cracking of the polycrystalline NCA particles.

2. Materials and methods

2.1. Electrode preparation

Polycrystalline NCA and LPSCI were supplied by Samsung SDI Co., LTD., Republic of Korea. Composite cathodes were prepared by mixing NCA, LPSCI and carbon black (Super P) powders using a ball mill. The weight ratio of NCA : LPSCI : Super P was 72:27:1. The LTO (Sigma Aldrich) composite anodes were fabricated by mixing LTO, LPSCI, and Super P in a weight ratio of 50:40:10. The morphological and structural features of NCA, LTO, and LPSCI used in this study are shown in Supplementary Note 1 (Figs. S1–S4).

2.2. Assembly of ASSBs

Cells were assembled using a poly(aryl-ether-ether-ketone) mold with a hole of 0.5 mm diameter for the reference electrode and stainless steel rods with a diameter of 10 mm. First, the LPSCI powder was added to a mold and pressed slightly. Subsequently, the composite cathode mixture was added onto the SE layer and pressed under 433 MPa, followed by attachment of In (thickness = 100 μm and diameter = 10 mm) and Li (thickness = 200 μm and diameter = 4 mm) foils to the other side of the SE layer. The mass loading of the cathode was 12.7 mg cm^{-2} . After inserting 10 μm -thick stainless steel foils (current collector) on both the electrodes, the entire cell was pressed under 250 MPa. To fabricate LTO cells, the LTO composite anode powder (mass loading of LTO = 20.4 mg cm^{-2} and N/P ratio = 1.66) was pressed under 433 MPa. The anode

stress/strain of the Li-In cell are exclusively determined by the capacity of the composite cathode, and thus, the N/P ratio has insignificant influence on the electrochemical performance degradation under our experimental conditions. Since the [cathode | SE] stacks of both cells were densified at the same maximum pressure of 433 MPa, the fabrication pressure of the anode is unlikely to have significant effects on the strain-induced cathode failure. Electrode preparation and cell assembly were performed inside an Ar-filled glove box to avoid exposure to humid air.

2.3. Electrochemical measurements

Cycling tests were conducted in the voltage range of 2.5–4.3 V vs. Li/Li^+ at 30°C. The initial cycling was performed at a 0.05C rate, where constant current-constant voltage (CC-CV) and constant current (CC) modes were used for charge and discharge, respectively. The cut-off current in the CV mode was set to 20 % of the applied current in the CC mode. After the initial cycle, the cells were charged and discharged at a 0.5C rate. Electrochemical impedance spectroscopy (EIS) tests were carried out using an impedance analyzer (Biologic SP-300) in the frequency range from 7 MHz to 10 mHz with an AC amplitude of 5 mV. For in-operando pressure measurements, pressure sensors with a resolution of 0.01 MPa (load cell, Bongshin) were assembled with the cells.

2.4. Material characterization

The morphologies and microstructures of the powders and the cross-sectional surfaces of the composite electrodes were characterized by scanning electron microscopy (SEM, Hitachi S-4800), in conjunction with energy dispersive X-ray spectroscopy (EDS) for elemental mapping. To observe the morphology of the composite cathodes, the cells were disassembled and the composite cathodes were polished to remove any remaining SE powder. For the cross-sectional SEM measurements, the cathodes were polished with an Ar ion beam using a cross-sectional polisher (Hitachi ArBlade5000). For structural characterization, X-ray diffraction (XRD) measurements were carried out using an X-ray diffractometer (Horiba, Miniflex 600 Mini) with $\text{Cu K}\alpha$ radiation (1.541 Å) in a dry room. The chemical compositions were analyzed by X-ray photoelectron spectroscopy (XPS) using a monochromatic Al $\text{K}\alpha$ source (1486.6 eV) on an ESCALAB 250Xi instrument (Thermo Scientific).

2.5. Modeling

Finite element analysis was performed to investigate the volume change-induced stresses in the active materials. For the cathode, the mechanical stress caused by volume change during the charging (delithiation) of the NCA was considered. The anode expansion during charging was expressed as the z -axis displacement of the anode-solid electrolyte interfacial boundary in the model. The polycrystalline structure of the NCA, consisting of primary particles, was generated using the Voronoi tessellation method, and the diameter of the secondary particles was assumed to be 10 μm (Fig. S1a). The domain size of the composite cathode was set to 12.5 μm along the x - and y -axis directions, considering the volume fractions of NCA and LPSCI in the real electrode. The mechanical moduli of NCA and LPSCI are 175 and 22.1 GPa, respectively [39]. In this model, the diffusion-induced strain (ϵ_d) of NCA caused by lithiation/delithiation was calculated using the relation $\epsilon_d = \Omega/3 \times \Delta c$, where Ω is the partial molar volume calculated based on 5.9 % volume change in NCA, and c is the Li concentration. Further details on the model are provided in Supplementary Note 2.

3. Results

ASSB cells are equipped with external jigs to maintain the stack pressure during the tests. Therefore, it is difficult to determine the strain/volume changes of full cells using in-operando dilatometric

measurements or optical observations. To assess the evolution of internal stress within the cells, the ASSB assemblies were equipped with pressure sensors, as illustrated schematically in Fig. 1a. The pressure sensor was placed on top of the cathode | SE | anode stack, allowing for real-time monitoring of the variation in the internal pressure during cell operation. The ASSB cells were operated at an initial stack pressure of 250 MPa, which was achieved by tightening the bolts on the cell. Fig. 1b and c show the evolution of the stack pressure measured during the cycling of Li-In and LTO cells, respectively. The results indicate that the Li-In anode expands during charging and contracts during discharging, whereas the NCA composite cathode exhibits the opposite behavior [40, 41]. The Li-In cell exhibited a high-pressure excursion of ~ 3.8 MPa, while the LTO cell underwent a slight fluctuation in pressure (~ 0.2 MPa). Given that the zero-strain LTO composite anode should have a negligible influence on the measured stress value, the stresses generated by the volume changes in the Li-In anode and the NCA composite cathode during cycling were approximately 4.0 and 0.2 MPa, respectively. The higher stress level of the Li-In cell compared to that of the LTO cell is fully attributed to the volume change in the Li-In anode upon Li intercalation-deintercalation. This large discrepancy in the volume change may have additional implications for the overall performance of the cell.

Fig. 1d–g illustrate the electrochemical performance of full cells

cycled between 2.5 and 4.3 V vs. Li/Li⁺ at 30°C. Following an initial cycle at a 0.05C rate, both cells underwent cycling at a current density of 0.5C. The Li-In and LTO cells exhibited similar discharge capacities of 152.7 and 152.4 mAh g⁻¹, respectively. For 100 cycles, no notable differences were observed between the two cells in terms of capacity retention (82.5 % for the Li-In cell and 81.6 % for the LTO cell). However, the Li-In cell experienced severe capacity loss beyond 120 cycles, and the specific capacity decreased dramatically to ~ 9 mAh g⁻¹ at the 200th cycle (Fig. 1d and e). In contrast, the LTO cell showed excellent cyclability for 200 cycles, with a capacity retention of 75.6 % and Coulombic efficiency of 99.97 % (Fig. 1f and g).

To probe the primary causes of the deterioration of the composite cathodes during cycling, we constructed three-electrode cells (Fig. S6) with a reference electrode (lithiated Ag wire) and acquired the AC-impedance spectra for each anode and cathode, as well as the data of the full cells (Fig. S7). The impedance spectra of the Li-In and LTO anodes showed no significant changes over 200 cycles (Fig. S8), indicating that the anodes were barely responsible for the cell degradation shown in Fig. 1d–g. The impedance spectrum of the LTO anode showed a slightly larger semi-circle after 50 cycles compared to that at the initial state, due probably to the rearrangement of constituent particles under pressure and stabilization of LTO/LPSCl interfaces with large contact areas; however, the anode quickly regained its stability. The stability of

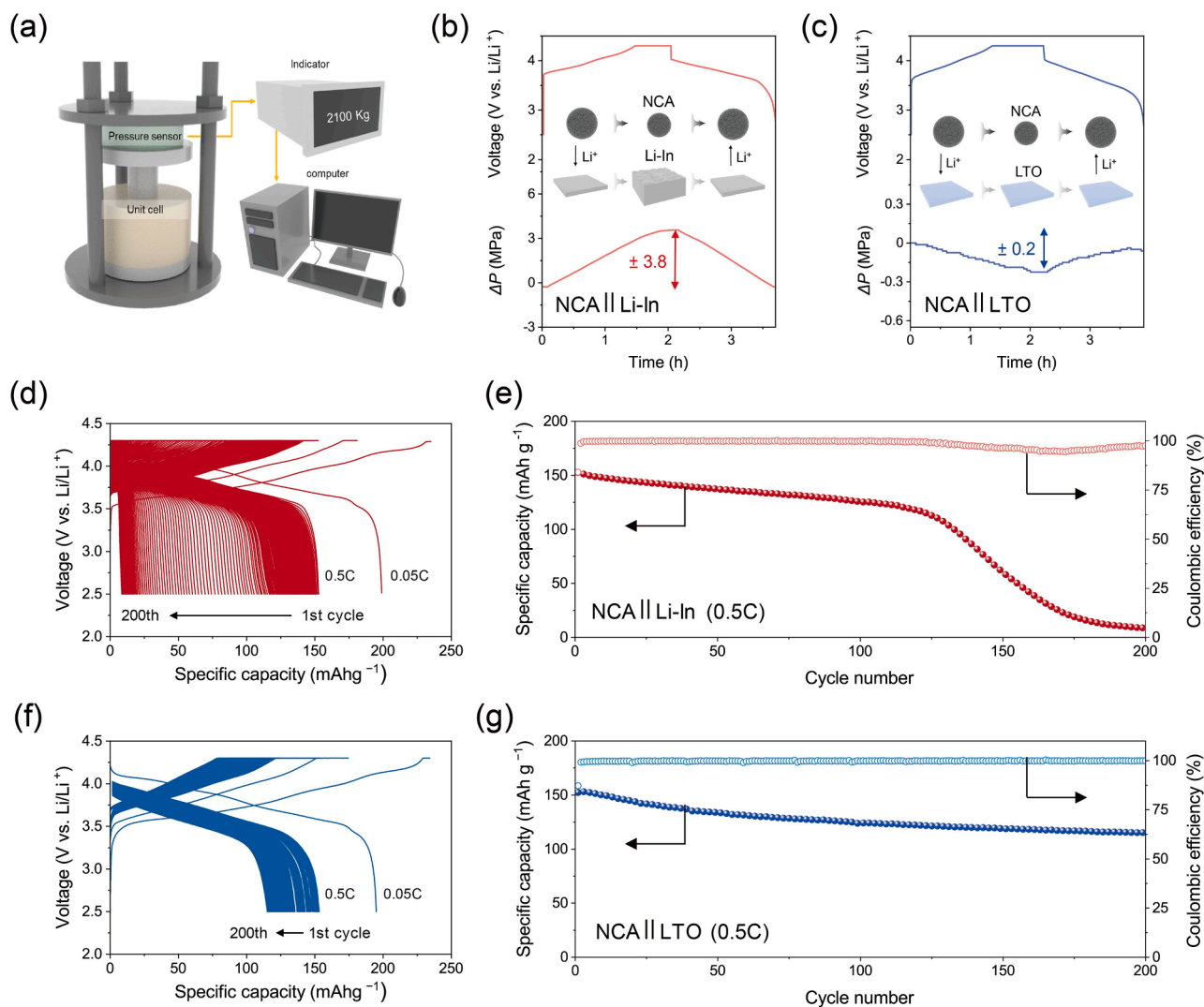


Fig. 1. (a) Schematic illustration of the cell assembly for *in situ* pressure measurement during cycling. *In situ* pressure measurement curves and voltage profiles of the (b) Li-In cell and (c) LTO cell. Cycling performances of the ASSBs with the (d, e) Li-In and (f, g) LTO anodes.

the Li-In anode in LPSCl was further confirmed through a symmetric [Li-In | LPSCl | Li-In] cell experiment (Fig. S9). The symmetric cells at a current density of 0.9 mA cm^{-2} (corresponding to 0.5C rate for full cells) showed stable cycling performance for 600 h (150 cycles). Furthermore, the variation in the ohmic resistance upon cycling (Fig. S10) confirmed that the SE layer did not influence the degradation behavior of the cells substantially.

The impedance signals of the NCA cathodes (after pre-conditioning) measured from the Li-In and LTO cells every 50 cycles are shown in Fig. 2a and b, respectively. Both cathodes from the as-prepared cells exhibited similar impedance behavior before pre-conditioning (Fig. S11); however, the cathode impedance of the Li-In cell increased more rapidly than that of the LTO cell after 100 cycles. Fig. 2c illustrates an equivalent circuit describing the multiple processes involved in a solid composite cathode based on a modified transmission-line model (TLM) [22,42,43,44]. The TLM used in this study was expressed by a network of distributed impedance elements to account for Li^+ transport, interfacial reactions at the NCA/SE interface, and solid-state diffusion inside the NCA particles. r_{ion} and r_{ele} denote the resistances per unit length for Li^+ and electron transport, respectively. The interfacial impedance comprises r_{ion} , cpe_{int} , and z_w , which denote the interfacial resistance, constant-phase element, and Warburg impedance, respectively. The impedance spectra of the NCA composite cathodes were fitted to the modified TLM, and the resulting parameters are summarized in Fig. 2d–f and Table S3. The ionic resistance (R_{ion}) values of the cathodes in the two cells were similar, suggesting that the ion-conducting properties had no significant impact on the rapid degradation observed for the Li-In cell (Fig. 1e). On the other hand, the interfacial resistance (R_{int}) and diffusion (Warburg) resistance (R_w) of the LTO cell increased slightly with cycle number, whereas these resistances increased dramatically for the Li-In cell after 150 cycles. Such steep changes in R_{int} ($65.2 \rightarrow 115.7 \Omega \text{ cm}^2$) and R_w ($24.4 \rightarrow 128.6 \Omega \text{ cm}^2$) from the 100th to 150th cycle may be closely related to the abrupt capacity decay of the Li-In cell at around the 120th cycle.

Further chemical and electrochemical experiments were conducted to determine the impact of anode volume changes on the composite cathodes. Fig. 3a and b show the XPS spectra of the S 2p and P 2p core levels in the composite cathodes after the initial and 200th cycles, respectively. Each spectrum was collected after charging the cell to 4.3 V vs. Li/Li^+ . The XPS peaks at 161.2 and 162.3 eV for S 2p corresponded to the PS_4^{3-} unit of LPSCl, while the newly generated signals at 162.7 and 163.4 eV were assigned to Li_2S_n ($n > 1$) and P_2S_x ($x > 5$), respectively [45,46]. The LPSCl phase in the LTO-cell maintained its original argyrodite structure after the initial cycle, whereas Li_2S_n and P_2S_x were formed in the Li-In cell. Li_2S_n is formed by the direct oxidation of LPSCl, while P_2S_x originates from the bridging of sulfur atoms upon oxidation, causing the positive shift of the binding energy due to the resultant S...S chains [47]. Because the formation of Li_2S_n and P_2S_x is related to oxidative side reactions at the SE/NCA interface and the XPS spectra showed no sign of LPSCl decomposition before pre-conditioning (Fig. S12), it is likely that upon the charging steps of the pre-conditioning cycle, the relatively large volume expansion of Li-In may have caused more intimate SE/NCA contacts (i.e., SE particles have more chance of interacting with NCA particles), leading to the further formation of chemical degradation products during the initial charge-discharge-charge cycle. After 200 cycles, the Li-In cell showed a much larger amount of P_2S_x (163.4 eV) than the LTO cell [47]. The PO_4^{3-} peak (134.1 eV) in Fig. 3a was also attributed to the oxidized species formed by electrolyte decomposition at the SE/NCA interface [48,49]. When the cell assembly was pressed again at 250 MPa after 150 cycles to restore the interfacial contact, the discharge capacity of the cell increased slightly from 25.2 to 38.3 mAh g^{-1} (Fig. S13), accompanied by a small decrease in R_{int} (Fig. S14). However, the capacity decay curve did not recover its original slope, indicating that the observed performance degradation was mainly due to irreversible processes rather than interfacial contact losses.

Thus, the cross-sections of the composite cathodes were examined by SEM (Fig. 4 and Fig. S15). Upon charging to 4.3 V vs. Li/Li^+ , several

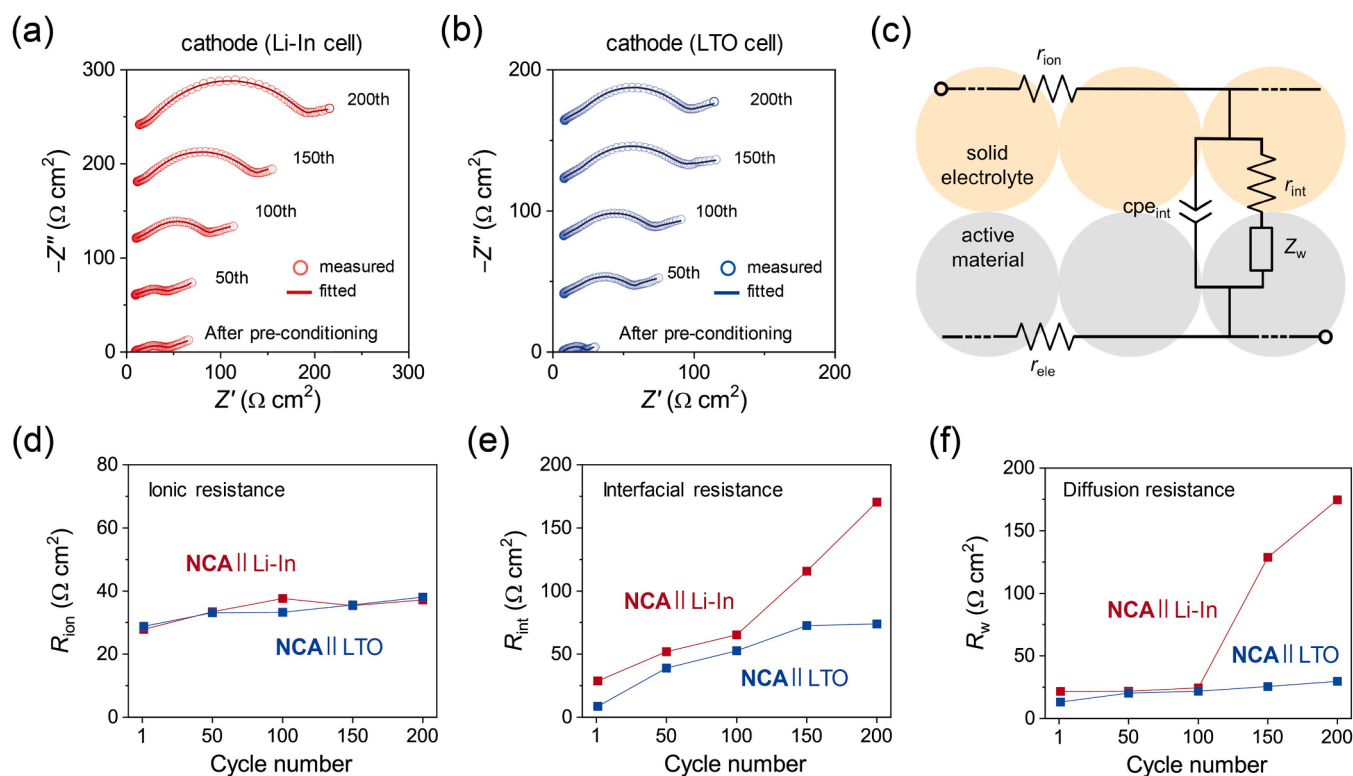


Fig. 2. AC-impedance spectra of the composite cathodes in the (a) Li-In and (b) LTO cells recorded after every 50 cycles. (c) Equivalent circuit model based on TLM. Area-specific (d) ionic, (e) interfacial, and (f) diffusion resistances ($\Omega \text{ cm}^2$) obtained by fitting the corresponding Nyquist plots in (a) and (b) to the TLM in (c).

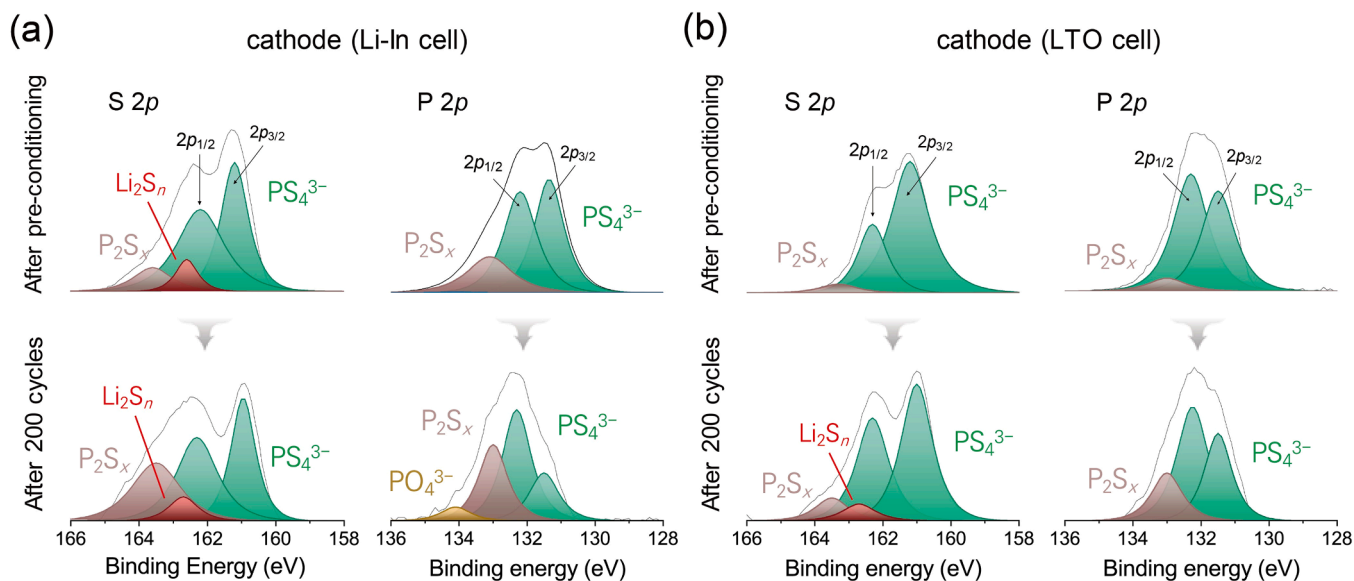


Fig. 3. XPS spectra for S 2p and P 2p core levels obtained from the cathodes of the (a) Li-In cell and (b) LTO cell before and after cycling.

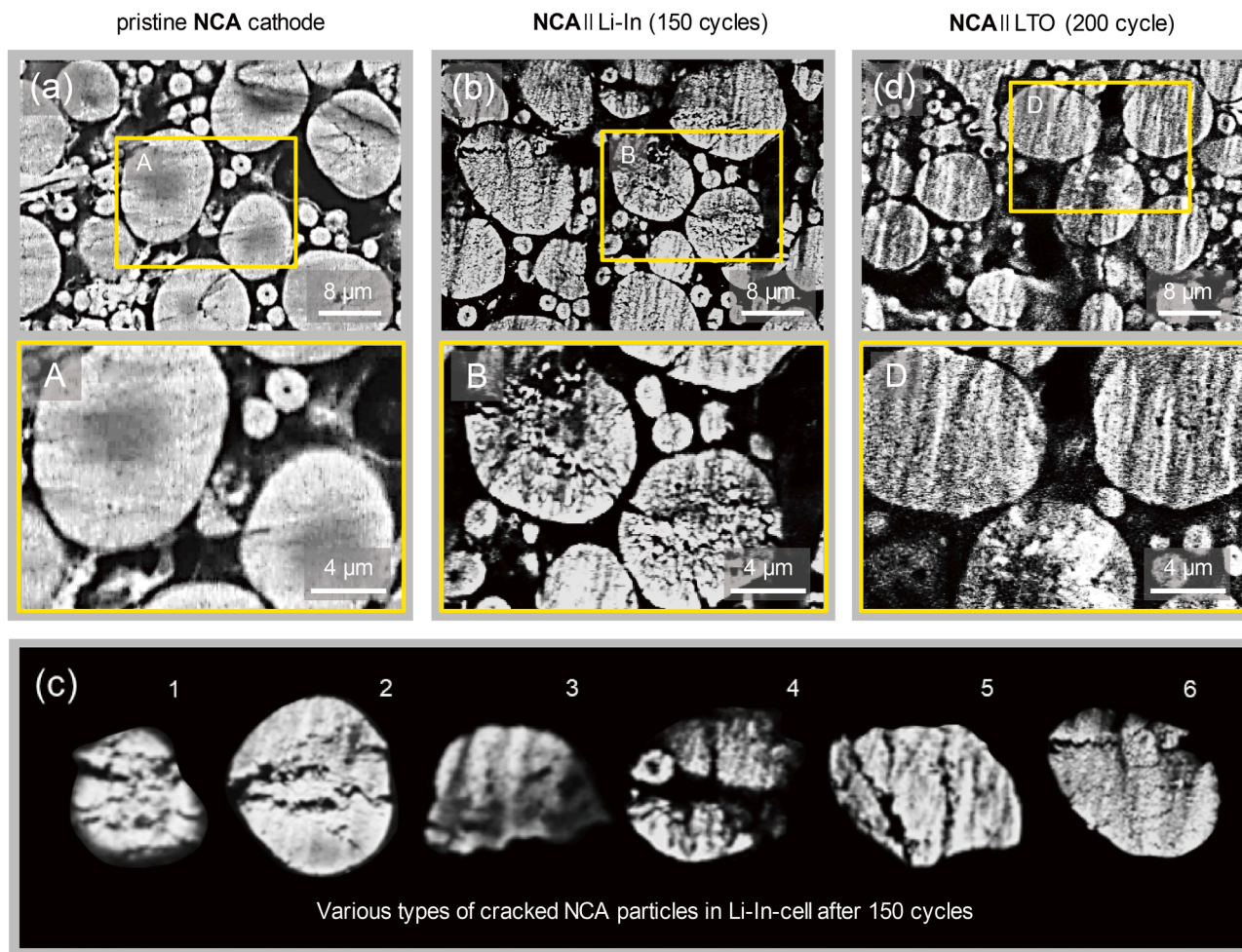


Fig. 4. Cross-sectional SEM images of the cathodes collected (a) before and (b-d) after cycling: (b, c) Li-In cell and (d) LTO cell.

microcracks appeared along the grain boundaries of the polycrystalline NCA particles, which could be attributed to the outward diffusion of Li^+ [50]. Notably, the NCA particles within the LTO cell remained nearly

intact even after 200 cycles (Fig. 4d), whereas those in the Li-In cell experienced significant cracking and pulverization after 150 cycles (Fig. 4b and c). The various types of cracked particles in the composite

cathode of the Li-In cell after cycling are shown in Fig. 4c. These images clearly indicate that the polycrystalline NCA particles are highly susceptible to cracking when coupled with a mechanically unstable anode.

4. Discussion

The electrochemical and post-mortem analyses provided clear evidence for the high-strain anode-induced deterioration of the NCA composite cathodes, based on the following: (i) the Li-In cells exhibited much faster deterioration than the LTO cells (Fig. 1); (ii) in situ impedance measurements indicated rapid increase in the interfacial and diffusion resistances of the composite cathodes (Fig. 2); (iii) XPS experiments demonstrated an increased amount of resistive interphases (for example, P_2S_x , Li_2S , and PO_4^{3-}) (Fig. 3); and (iv) cross-sectional SEM images indicated severe cracking of the polycrystalline NCA particles (Fig. 4). Based on our experimental findings, Fig. 5 schematically illustrates how the volume changes in the high-strain anode can be linked to the chemo-mechanical degradation of the solid composite cathode. Compared to the zero-strain anode, the composite cathode is subjected to repeated (cyclic) tension-compression loads at higher levels because of the high-strain anode. That is, the high-strain anode perturbs the structural integrity of the composite cathode and facilitates “dynamic” contacts (strongly adhered, weakly adhered, newly exposed) among the cathode constituents during cycling. Such dynamic contact conditions increase the amount of the parasitic reaction products with diverse chemical compositions and concentrations (i.e., interfacial heterogeneity) on the outer NCA surfaces, resulting in the progressive degradation of the composite cathode in the early stage of cycling (< 120 cycles). This interfacial heterogeneity generates localized concentrations of diffusion-induced stress at the GPa level in the NCA, which, in turn, accelerates the cracking of the NCA aggregates and causes the disruptive degradation in the later stage of cycling (> 120 cycles) [51,52]. Although SEs are less likely to penetrate into the cracked NCA particles, the fracture of the NCA active materials further exacerbates the aforementioned interfacial instability by providing additional reaction sites on the outer NCA surfaces, initiating a detrimental cycle of battery degradation. The fracture of NCA particles leads to the formation of electrically disconnected, isolated fragments and thus reduces the electrochemically active surface areas while making the diffusion pathways tortuous. This was evidenced by drastic increases in R_{int} and R_w for the Li-In cell after 100 cycles (Fig. 2). In this regard, Ni-rich oxide cathode materials, which undergo much larger volume contraction upon charging than NCM cathodes with low a Ni content and $LiCoO_2$ cathodes

[53,54], would be more sensitive to anode-derived degradation.

Finally, it should be noted that the high-strain anode can exert a large stress on the cell components, causing extensive mechanical damage to the NCA particles. Thus, we employed electrochemical-mechanical modeling to investigate the influence of anode-induced stress and interfacial heterogeneity on the fracture of polycrystalline the NCA particles. Along with the volume contraction during the delithiation of the NCA particles, we considered the compressive stress (σ_{anode}) caused by the expansion of the anode and the applied stack pressure (σ_{ext}). A series of numerical tests were performed by using the σ_{anode} values measured at various states of charge (SOC) and the surface coverages of the resistive phases to identify the influence of anode expansion on NCA particles (Fig. 6 for stress distributions and Fig. S16 for Li concentration distributions). The von Mises stress (σ_{Mises}), which is used to determine whether a given material will yield or fail, is an effective parameter for assessing the probability of plastic deformation of active material particles. As the charging progressed, the Li-In cell under our experimental conditions underwent a more rapid increase in σ_{Mises} along the grain boundaries, compared to the LTO cell. Specifically, intergranular fracture was the dominant damage mode that degraded the connectivity of the subsurface grains within the secondary aggregate. These simulation results demonstrate that the anode-derived stress combined with interfacial heterogeneity cause substantial mechanical damages of polycrystalline particles, which ultimately accelerate the cell degradation.

In summary, we demonstrated that high-strain anodes can accelerate the chemo-mechanical degradation of composite cathodes in ASSBs. A series of chemical, electrochemical, and microstructural analyses suggest that the high-strain anode induces dynamic contacts between the cathode constituents upon repeated cycling, rendering the NCA surfaces electrochemically heterogeneous. This eventually leads to the cracking of the NCA particles and is the main cause of rapid performance decay. The electrochemical-mechanical modeling results also support the accelerated mechanical failure of the NCA particles due to the high-strain anodes. It should be noted that the performance degradation of the composite cathodes is strongly dependent on various factors, such as the areal capacity, applied stack pressure, active material (composition and particle type – poly- or single-crystalline), and electrode formulation. For example, the Li-In cell cycled at a reduced stack pressure (100 MPa) maintained stable cyclability over 300 cycles (Fig. S17). The role of high-strain anodes in enhancing the cathode degradation may be operative in liquid-electrolyte-based batteries, e.g., Li-metal batteries; however, liquid electrolytes can buffer the anode-derived stress to some extent, thus reducing the chemo-mechanical degradation of cathodes

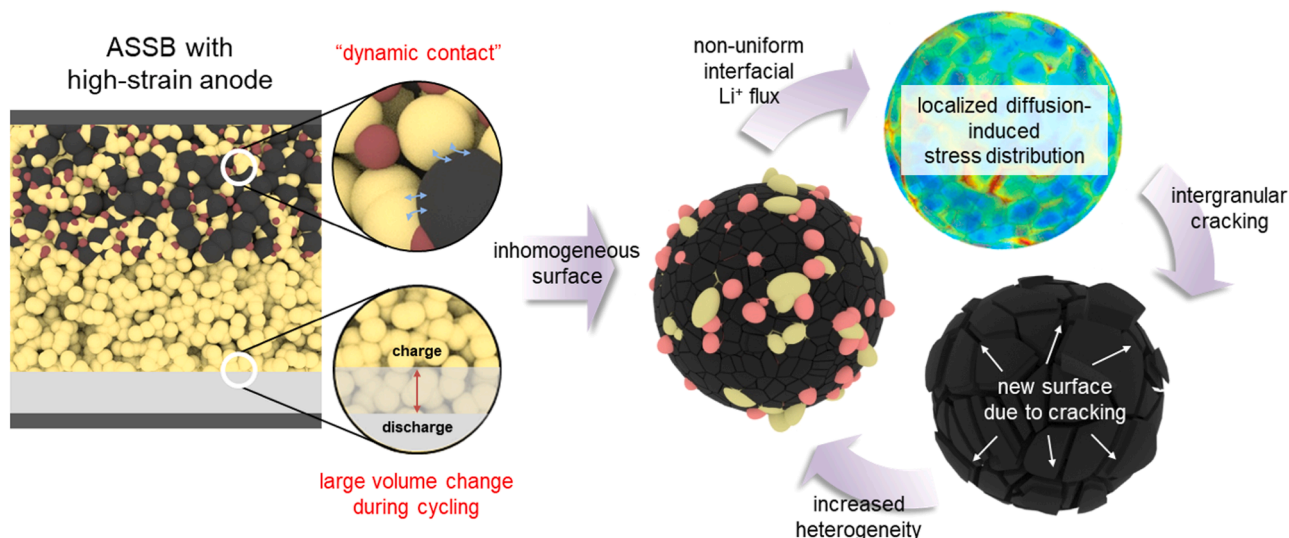


Fig. 5. Schematic illustration of the chemo-mechanical degradation of the cathode caused by the high-strain anode during cycling.

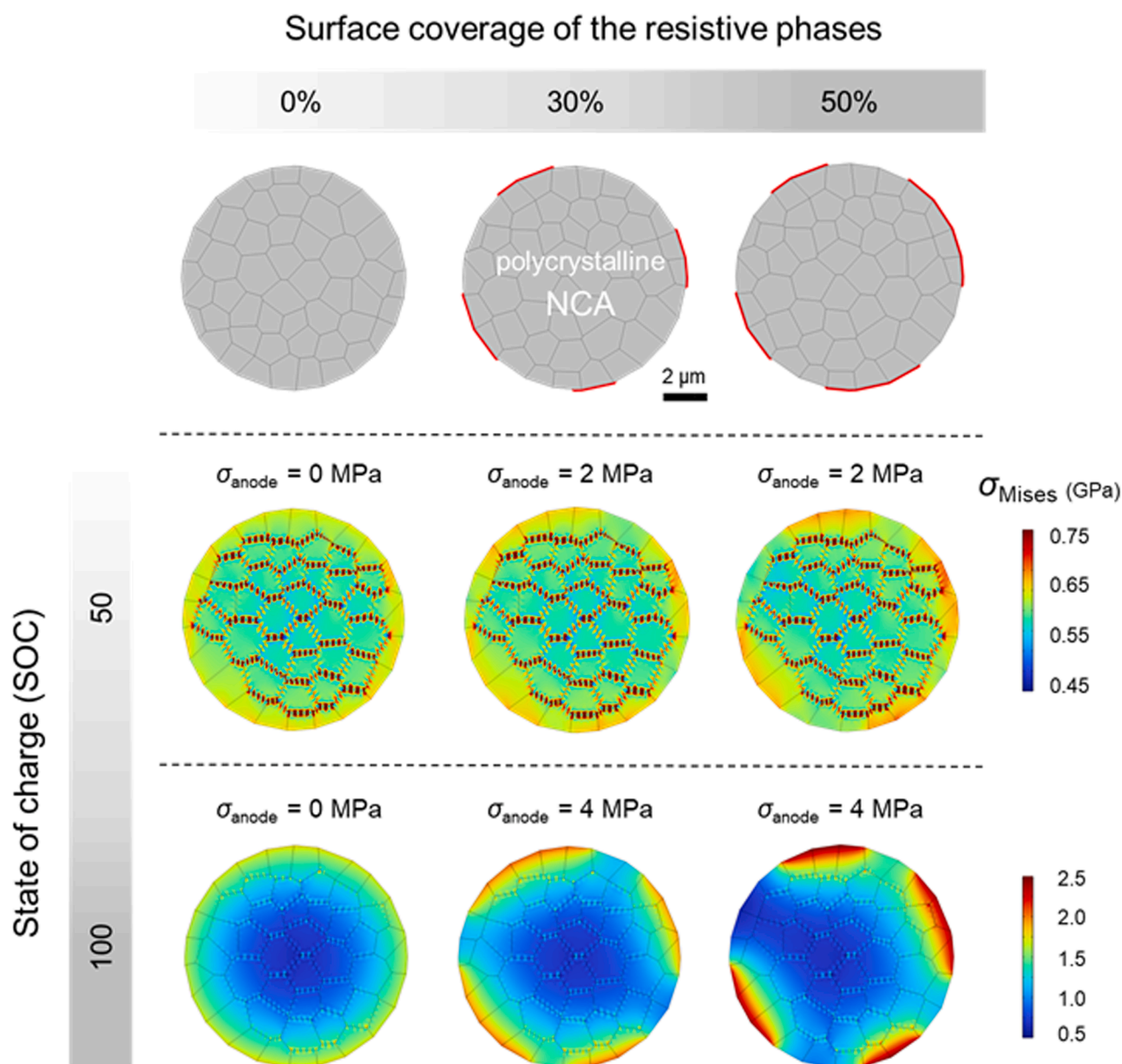


Fig. 6. Distribution of von Mises stress in the polycrystalline NCA particle upon charging simulated in consideration of the anode-derived stress and interfacial heterogeneity.

due to high-strain anodes. Given that the chemo-mechanical degradation of the cathode is accelerated by the fluctuation in cell pressure, real-time pressure regulation can be helpful for stable cycling, e.g., spring-equipped ASSB assembly [55]. Furthermore, the design and construction of ASSBs with low-strain cathode materials and SE materials capable of accommodating mechanical stresses can minimize the cell degradation due to high-strain anodes. This study provides insights into the influence of high-strain anodes on the accelerated degradation of composite cathodes and will contribute to the development of sulfide-based ASSBs with a long cycle life.

CRediT authorship contribution statement

Junhee Kang: Investigation, Writing – original draft. **Hong Rim Shin:** Formal analysis. **Jonghyeok Yun:** Investigation. **Siwon Kim:** Formal analysis. **Beomsu Kim:** Investigation. **Kyeongsu Lee:** Investigation. **Youngjun Lim:** Investigation. **Jong-Won Lee:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by Samsung SDI Co., LTD., National Research Foundation of Korea (2018R1A5A1025594), and Korea Evaluation Institute of Industrial Technology (20010044).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ensm.2023.103049](https://doi.org/10.1016/j.ensm.2023.103049).

References

- [1] Y. Liang, C.Z. Zhao, H. Yuan, Y. Chen, W. Zhang, J.Q. Huang, D. Yu, Y. Liu, M. M. Titirici, Y.L. Chueh, H. Yu, Q. Zhang, A review of rechargeable batteries for portable electronic devices, *InfoMat* 1 (2019) 6–32, <https://doi.org/10.1002/inf2.12000>.
- [2] X. Zhang, J. Chen, Z. Xu, Q. Dong, H. Ao, Z. Hou, Y. Qian, Aqueous electrolyte with moderate concentration enables high-energy aqueous rechargeable lithium ion battery for large scale energy storage, *Energy Storage Mater.* 46 (2022) 147–154, <https://doi.org/10.1016/j.ensm.2022.01.009>.
- [3] S. Akhtar, W. Lee, M. Kim, M.S. Park, W.S. Yoon, Conduction mechanism of charge carriers in electrodes and design factors for the improvement of charge conduction in Li-ion batteries, *J. Electrochem. Sci. Technol.* 12 (2021) 1–20, <https://doi.org/10.33961/jecst.2020.01564>.
- [4] M.M. Thackeray, C. Wolverton, E.D. Isaacs, Electrical energy storage for transportation—approaching the limits of, and going, beyond, lithium-ion batteries, *Energy Environ. Sci.* 5 (2012) 7854–7863, <https://doi.org/10.1039/C2EE21892E>.
- [5] J.W. Choi, D. Aurbach, Promise and reality of post-lithium-ion batteries with high energy densities, *Nat. Rev. Mater.* 1 (2016) 16013, <https://doi.org/10.1038/natrevmats.2016.13>.
- [6] Y. Choi, J. Moon, J. Yun, K.N. Jung, J.W. Moon, J.W. Lee, Liquid electrolyte-free cathode for long-cycle life lithium-oxygen batteries, *Chem. Eng. J.* 420 (2021), 129840, <https://doi.org/10.1016/j.cej.2021.129840>.
- [7] Y. Chen, Y. Kang, Y. Zhao, L. Wang, J. Liu, Y. Li, Z. Liang, X. He, X. Li, N. Tavajohi, B. Li, A review of lithium-ion battery safety concerns: the issues, strategies, and testing standards, *J. Energy Chem.* 59 (2021) 83–99, <https://doi.org/10.1016/j.jchem.2020.10.017>.
- [8] S. Chen, K. Wen, J. Fan, Y. Bando, D. Golberg, Progress and future prospects of high-voltage and high-safety electrolytes in advanced lithium batteries: from liquid to solid electrolytes, *J. Mater. Chem. A* 6 (2018) 11631–11663, <https://doi.org/10.1039/C8TA03358G>.
- [9] P. Sun, R. Bisschop, H. Niu, X. Huang, A review of battery fires in electric vehicles, *Fire Technol.* 56 (2020) 1361–1410, <https://doi.org/10.1007/s10694-019-00944-3>.
- [10] Y.K. Sun, Promising all-solid-state batteries for future electric vehicles, *ACS Energy Lett.* 5 (2020) 3221–3223, <https://doi.org/10.1021/acseenergylett.0c01977>.
- [11] W. Zhao, J. Yi, P. He, H. Zhou, Solid-state electrolytes for lithium-ion batteries: fundamentals, challenges and perspectives, *Electrochem. Energy Rev.* 2 (2019) 574–605, <https://doi.org/10.1007/s41918-019-00048-0>.
- [12] E.S. Won, H.R. Shin, W. Jeong, J. Yun, J.W. Lee, Biphasic solid electrolytes with homogenous Li-ion pathway enabled by metal-organic frameworks, *Electrochim. Acta* 418 (2022), 140374, <https://doi.org/10.1016/j.electacta.2022.140374>.
- [13] W. Jeong, S.S. Park, J. Yun, H.R. Shin, J. Moon, J.W. Lee, Tailoring grain boundary structures and chemistry of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ solid electrolytes for enhanced air stability, *Energy Storage Mater.* 54 (2023) 543–552, <https://doi.org/10.1016/j.ensm.2022.10.044>.
- [14] K.N. Jung, H.S. Shin, M.S. Park, J.W. Lee, Solid-state lithium batteries: bipolar design, fabrication, and electrochemistry, *ChemElectroChem* 6 (2019) 3842–3859, <https://doi.org/10.1002/celec.201900736>.
- [15] X. Gao, X. Yang, M. Jiang, M. Zheng, Y. Zhao, R. Li, W. Ren, H. Huang, R. Sun, J. Wang, C.V. Singh, X. Sun, Fast ion transport in Li-rich alloy anode for high-energy-density all solid-state lithium metal batteries, *Adv. Funct. Mater.* 33 (2023), 2209715, <https://doi.org/10.1002/adfm.202209715>.
- [16] G.H. Lee, S.G. Lee, S.H. Park, D. Jun, Y.J. Lee, Interface engineering on a Li metal anode for an electro-chemo-mechanically stable anodic interface in all-solid-state batteries, *J. Mater. Chem. A* 10 (2022) 10662–10671, <https://doi.org/10.1039/D1TA10666G>.
- [17] J.A. Lewis, K.A. Cavallaro, Y. Liu, M.T. McDowell, The promise of alloy anodes for solid-state batteries, *Joule* 6 (2022) 1418–1430, <https://doi.org/10.1016/j.joule.2022.05.016>.
- [18] D. Cao, X. Sun, Y. Wang, H. Zhu, Bipolar stackings high voltage and high cell level energy density sulfide based all-solid-state batteries, *Energy Storage Mater.* 48 (2022) 458–465, <https://doi.org/10.1016/j.ensm.2022.03.012>.
- [19] H.S. Shin, W. Jeong, M.H. Ryu, S.W. Lee, K.N. Jung, J.W. Lee, Electrode-to-electrode monolithic integration for high-voltage bipolar solid-state batteries based on plastic-crystal polymer electrolyte, *Chem. Eng. J.* 433 (2022), 133753, <https://doi.org/10.1016/j.cej.2021.133753>.
- [20] J. Lau, R.H. DeBlock, D.M. Butts, D.S. Ashby, C. C.S., B.S. Dunn, Sulfide solid electrolytes for lithium battery applications, *Adv. Energy Mater.* 8 (2018), 1800933, <https://doi.org/10.1002/aenm.201800933>.
- [21] S. Chen, D. Xie, G. Liu, J.P. Mwizerwa, Q. Zhang, Y. Zhao, X. Xu, X. Yao, Sulfide solid electrolytes for all-solid-state lithium batteries: Structure, conductivity, stability and application, *Energy Storage Mater.* 14 (2018) 58–74, <https://doi.org/10.1016/j.ensm.2018.02.020>.
- [22] J. Yun, H.R. Shin, T.D. Hoang, S. Kim, J.H. Choi, B. Kim, H. Jung, J. Moon, J. W. Lee, Deciphering the critical degradation factors of solid composite electrodes with halide electrolytes: interfacial reaction versus ionic transport, *Energy Storage Mater.* 59 (2023), 102787, <https://doi.org/10.1016/j.ensm.2023.102787>.
- [23] K.J. Kim, M. Balaish, M. Wadaguchi, L. Kong, J.L.M. Rupp, Solid-state Li-metal batteries: challenges and horizons of oxide and sulfide solid electrolytes and their interfaces, *Adv. Energy Mater.* 11 (2021), 2022689, <https://doi.org/10.1002/aenm.202002689>.
- [24] W. Xu, J. Wang, F. Ding, X. Chen, E. Nasybulin, Y. Zhang, J.G. Zhang, Lithium metal anodes for rechargeable batteries, *Energy Environ. Sci.* 7 (2014) 513–537, <https://doi.org/10.1039/C3EE40795K>.
- [25] X.B. Cheng, R. Zhang, C.Z. Zhao, Q. Zhang, Toward safe lithium metal anode in rechargeable batteries: a review, *Chem. Rev.* 117 (2017) 10403–10473, <https://doi.org/10.1021/acs.chemrev.7b00115>.
- [26] Y.G. Lee, S. Fujiki, C. Jung, N. Suzuki, N. Yashiro, R. Omoda, D.S. Ko, T. Shiratsuchi, T. Sugimoto, S. Ryu, J.H. Ku, T. Watanabe, Y. Park, U. Aihara, D. Im, I.T. Han, High-energy long-cycling all-solid-state lithium metal batteries enabled by silver–carbon composite anodes, *Nat. Energy* 5 (2020) 299–308, <https://doi.org/10.1038/s41560-020-0575-z>.
- [27] Y. Huang, B. Shao, F. Han, Li alloy anodes for high-rate and high-areal-capacity solid-state batteries, *J. Mater. Chem. A* 10 (2022) 12350–12358, <https://doi.org/10.1039/D2TA02339C>.
- [28] X. Zhan, M. Li, S. Li, X. Pang, F. Mao, H. Wang, Z. Sun, X. Han, B. Jiang, Y.B. He, M. Li, Q. Zhang, Challenges and opportunities towards silicon-based all-solid-state batteries, *Energy Storage Mater.* 61 (2023), 102875, <https://doi.org/10.1016/j.ensm.2023.102875>.
- [29] S.Y. Han, C. Lee, J.A. Lewis, D. Yeh, Y. Liu, H.W. Lee, M.T. McDowell, Stress evolution during cycling of alloy-anode solid-state batteries, *Joule* 5 (2021) 2450–2465, <https://doi.org/10.1016/j.joule.2021.07.002>.
- [30] T. Shi, Y.Q. Zhang, Q. Tu, Y. Wang, M.C. Scott, G. Ceder, Characterization of mechanical degradation in an all-solid-state battery cathode, *J. Mater. Chem. A* 8 (2020) 17399–17404, <https://doi.org/10.1039/D0TA06985J>.
- [31] Y. He, C. Lu, S. Liu, W. Zheng, J. Luo, Interfacial incompatibility and internal stresses in all-solid-state batteries, *Adv. Energy Mater.* 9 (2019), 1901810, <https://doi.org/10.1002/aenm.201901810>.
- [32] G. Bucci, T. Swamy, Y.M. Chiang, W.C. Carter, Modeling of internal mechanical failure of all-solid-state batteries during electrochemical cycling, and implications for battery design, *J. Mater. Chem. A* 5 (2017) 19422–19430, <https://doi.org/10.1039/c7ta03199h>.
- [33] W.J. Zhang, A review of the electrochemical performance of alloy anodes for lithium-ion batteries, *J. Power Sources* 196 (2011) 13–24, <https://doi.org/10.1016/j.jpowsour.2010.07.020>.
- [34] Y. Liang, H. Liu, G. Wang, C. Wang, Y. Ni, C.W. Nan, L.Z. Fan, Challenges, interface engineering, and processing strategies toward practical sulfide-based all-solid-state lithium batteries, *InfoMat* 4 (2022) e12292, <https://doi.org/10.1002/inf2.12292>.
- [35] J. Gu, Z. Liang, J. Shi, Y. Yang, Electrochemo-mechanical stresses and their measurements in sulfide-based all-solid-state batteries, *Adv. Energy Mater.* 13 (2023), 2203153, <https://doi.org/10.1002/aenm.202203153>.
- [36] S. Xia, X. Wu, Z. Zhang, Y. Cui, X. Liu, Practical challenges and future perspectives of all-solid-state lithium-metal batteries, *Chem.* 5 (2019) 753–785, <https://doi.org/10.1016/j.chempr.2018.11.013>.
- [37] B.A. Budiman, A. Saputro, S. Rahardian, M. Aziz, P. Sambergoro, I.P. Nurprasetyo, Mechanical damages in solid electrolyte battery due to electrode volume changes, *J. Energy Storage* 52 (2022), 104810, <https://doi.org/10.1016/j.jest.2022.104810>.
- [38] X. Wang, K. He, S. Li, J. Zhang, Y. Lu, Realizing high-performance all-solid-state batteries with sulfide solid electrolyte and silicon anode: a review, *Nano Res.* 16 (2023) 3741–3765, <https://doi.org/10.1007/s12274-022-4526-9>.
- [39] R. Koerver, W. Zhang, L. Biasi, S. Schweidler, A.O. Kondrakov, S. Kolling, T. Brezesinski, P. Hartmann, W.G. Zeier, J. Janek, Chemo-mechanical expansion of lithium electrode materials – on the route to mechanically optimized all-solid-state batteries, *Energy Environ. Sci.* 11 (2018) 2142–2158, <https://doi.org/10.1039/C8EE00907D>.
- [40] B. Li, Z. Sun, N. Lv, Y. Hu, L. Jiang, Z. Zhang, F. Liu, Dual protection of a Li–Ag alloy anode for all-solid-state lithium batteries with the argyrodite $\text{Li}_6\text{PS}_5\text{Cl}$ solid electrolyte, *ACS Appl. Mater. Interfaces* 14 (2022) 37738–37746, <https://doi.org/10.1021/acsaami.2c09013>.
- [41] A. Banerjee, H. Tang, X. Wang, J.H. Cheng, H. Nguyen, M. Zhang, D.H.S. Tan, T. A. Wynn, E.A. Wu, J.M. Dour, T. Wu, L. Ma, G.E. Sterbinsky, M.S. D’Souza, S. P. Ong, Y.S. Meng, Revealing nanoscale solid-solid interfacial phenomena for long-life and high-energy all-solid-state batteries, *ACS Appl. Mater. Interfaces* 11 (2019) 43138–43145, <https://doi.org/10.1021/acsaami.9b13955>.
- [42] S. Larfaillou, D. Guy-Bouysou, F. Cras, S. Franger, Comprehensive characterization of all-solid-state thin films commercial microbatteries by electrochemical impedance spectroscopy, *J. Power Sources* 319 (2016) 139–146, <https://doi.org/10.1016/j.jpowsour.2016.04.057>.
- [43] S.K. Jung, H. Gwon, S.S. Lee, H. Kim, J.C. Lee, J.G. Chung, S.Y. Park, Y. Aihara, D. Im, Understanding the effects of chemical reactions at the cathode–electrolyte interface in sulfide based all-solid-state batteries, *J. Mater. Chem. A* 7 (2019) 22967–22976, <https://doi.org/10.1039/c9ta08517c>.
- [44] N. Kaiser, S. Spannenberger, M. Schmitt, M. Cronau, Y. Kato, B. Roling, Ion transport limitations in all-solid-state lithium battery electrodes containing a sulfide-based electrolyte, *J. Power Sources* 396 (2018) 175–181, <https://doi.org/10.1016/j.jpowsour.2018.05.095>.
- [45] F. Walther, R. Koerver, T. Fuchs, S. Ohno, J. Sann, M. Rohnke, W.G. Zier, J. Janek, Visualization of the interfacial decomposition of composite cathodes in argyrodite-based all-solid-state batteries using time-of-flight secondary-ion mass spectrometry, *Chem. Mater.* 31 (2019) 3745–3755, <https://doi.org/10.1021/acs.chemmater.9b00770>.
- [46] X. Sun, L. Wang, J. Ma, X. Yu, S. Zhang, X. Zhou, G. Cui, A bifunctional chemomechanics strategy to suppress electrochemo-mechanical failure of Ni-rich

- cathodes for all-solid-state lithium batteries, *ACS Appl. Mater. Interfaces* 14 (2022) 17674–17681, <https://doi.org/10.1021/acsami.2c02678>.
- [47] T. Rödl, R. Wehrich, J. Wack, J. Senker, A. Pfitzner, Rational syntheses and structural characterization of sulfur-rich phosphorus polysulfides: α -P₂S₇ and β -P₂S₇, *Angew. Chem.* 50 (2011) 10996–11000, <https://doi.org/10.1002/anie.201103485>.
- [48] T. Kim, K. Kim, S. Lee, G. Song, M.S. Jung, K.T. Lee, Thermal runaway behavior of Li₆PS₅Cl solid electrolytes for LiNi_{0.9}Co_{0.1}Mn_{0.1}O₂ and LiFePO₄ in all-solid-state batteries, *Chem. Mater.* 34 (2022) 9159–9171, <https://doi.org/10.1021/acs.chemmater.2c02106>.
- [49] A. Banerjee, H. Tang, X. Wang, J.H. Cheng, H. Nguyen, M. Zhang, D.H.S. Tan, T. A. Wynn, E.A. Wu, J.M. Doux, T. Wu, L. Ma, G.E. Sterbinsky, M.S. D'Souza, S. P. Ong, Revealing nanoscale solid-solid interfacial phenomena for long-life and high-energy all-solid-state batteries, *ACS Appl. Mater. Interfaces* 11 (2019) 43138–43145, <https://doi.org/10.1021/acsami.9b13955>.
- [50] K.J. Park, J.Y. Hwang, H.H. Ryu, F. Maglia, S.J. Kim, P. Lamp, C.S. Yoon, Y.K. Sun, Degradation mechanism of Ni-enriched NCA cathode for lithium batteries: are microcracks really critical? *ACS Energy Lett.* 4 (2019) 1394–1400, <https://doi.org/10.1021/acsenergylett.9b00733>.
- [51] H.K. Tian, A. Chakraborty, A.A. Talin, P. Eisenlohr, Y. Qi, Evaluation of the electrochemo-mechanically induced stress in all-solid-state Li-ion batteries, *J. Electrochem. Soc.* 167 (2020), 090541, <https://doi.org/10.1149/1945-7111/ab8f5b>.
- [52] H. Liu, M. Wolfman, K. Karki, Y.S. Yu, E.A. Stach, J. Cabana, K.W. Chapman, P. J. Chupas, Intergranular cracking as a major cause of long-term capacity fading of layered cathodes, *Nano Lett.* 17 (2017) 3452–3457, <https://doi.org/10.1021/acs.nanolett.7b00379>.
- [53] A.O. Kondrakov, A. Schmidt, J. Xu, H. Geßwein, R. Mönig, P. Hartmann, H. Sommer, T. Brezesinski, J. Janek, Anisotropic lattice strain and mechanical degradation of high- and low-nickel NCM cathode materials for Li-ion batteries, *J. Phys. Chem. C* 121 (2017) 3286–3294, <https://doi.org/10.1021/acs.jpcc.6b12885>.
- [54] H.H. Ryu, K.J. Park, C.S. Yoon, Y.K. Sun, Capacity fading of Ni-rich Li[Ni_xCo_yMn_{1-x-y}]O₂ (0.6 ≤ x ≤ 0.95) cathodes for high-energy-density lithium-ion batteries: bulk or surface degradation? *Chem. Mater.* 30 (2018) 1155–1163, <https://doi.org/10.1021/acs.chemmater.7b05269>.
- [55] Y.S. Park, K. Kim, J.W. Lee, J.W. Moon, H.H. Park, J. Hwang, Effect of cell pressure on the electrochemical performance of all-solid-state lithium batteries with zero-excess Li metal anode, *J. Am. Ceram. Soc.* 106 (2023) 7322–7330, <https://doi.org/10.1111/jace.19322>.